

Animal.h Copy-paste code

```
#include <iostream>
#ifndef ANIMAL_H

class Animal {
public:
    std::string name, species;

    void display() { // Overridden in ZooAnimal.cpp
        std::cout << "Name:\t" << name << std::endl;
        std::cout << "Species:\t" << species << std::endl;
    }
};

class Habitat {
public:
    std::string type;
    double size;

    void display() { // Overridden in ZooAnimal.cpp
        std::cout << "Type:\t" << type << std::endl;
        std::cout << "Size:\t" << size << std::endl;
    }
};

#endif
```

ZooAnimal.cpp Copy-paste code

```
#include "Animal.h"

// Inherits from Animal and Habitat classes
class ZooAnimal : public Animal, public Habitat {
public:
    int id;
    std::string diet;

    void display() {
        std::cout << "Name: " << name << std::endl;
        std::cout << "Species: " << species << std::endl;
        std::cout << "Habitat: " << type << std::endl;
        std::cout << "Size: " << size << std::endl;
        std::cout << "ID: " << id << std::endl;
        std::cout << "Diet: " << diet << std::endl;
    }
};

int main() {
    ZooAnimal a1;
    a1.name = "Bob";
    a1.species = "Giraffe";
    a1.type = "Savannah";
    a1.size = 1000;
    a1.id = 1;
```

```
a1.diet = "Herbivore";

a1.display();
std::cout << std::endl;

ZooAnimal a2;
a2.name = "Clifford";
a2.species = "Big Red Dog";
a2.type = "House";
a2.size = 125;
a2.id = 2;
a2.diet = "Omnivore";

a2.display();

return 0;
}
```

Screenshots:

```
1  #include <iostream>
2  #ifndef ANIMAL_H
3
4  class Animal {
5  public:
6      std::string name, species;
7
8      void display() { // Overridden in ZooAnimal.cpp
9          std::cout << "Name:\t" << name << std::endl;
10         std::cout << "Species:\t" << species << std::endl;
11     }
12 };
13
14 class Habitat {
15 public:
16     std::string type;
17     double size;
18
19     void display() { // Overridden in ZooAnimal.cpp
20         std::cout << "Type:\t" << type << std::endl;
21         std::cout << "Size:\t" << size << std::endl;
22     }
23 };
24
25 #endif #ifndef ANIMAL_H
```

```
1  ▶ #include "Animal.h"
2
3  // Inherits from Animal and Habitat classes
4  class ZooAnimal : public Animal, public Habitat {
5  public:
6      int id;
7      std::string diet;
8
9  Ⓢ void display() {
10      std::cout << "Name: " << name << std::endl;
11      std::cout << "Species: " << species << std::endl;
12      std::cout << "Habitat: " << type << std::endl;
13      std::cout << "Size: " << size << std::endl;
14      std::cout << "ID: " << id << std::endl;
15      std::cout << "Diet: " << diet << std::endl;
16  }
17  };
18
```

```
19  ▶ int main() {  
20      ZooAnimal a1;  
21      a1.name = "Bob";  
22      a1.species = "Giraffe";  
23      a1.type = "Savannah";  
24      a1.size = 1000;  
25      a1.id = 1;  
26      a1.diet = "Herbivore";  
27  
28      a1.display();  
29      std::cout << std::endl;  
30  
31      ZooAnimal a2;  
32      a2.name = "Clifford";  
33      a2.species = "Big Red Dog";  
34      a2.type = "House";  
35      a2.size = 125;  
36      a2.id = 2;  
37      a2.diet = "Omnivore";  
38  
39      a2.display();  
40  
41      return 0;  
42  }
```

Output:

```
Name: Bob  
Species: Giraffe  
Habitat: Savannah  
Size: 1000  
ID: 1  
Diet: Herbivore
```

```
Name: Clifford  
Species: Big Red Dog  
Habitat: House  
Size: 125  
ID: 2  
Diet: Omnivore
```

```
Process finished with exit code 0
```